

・循環小数と無限等比級数

(例) 次の循環小数を分数に直せ。

(1) $0.\dot{1}\dot{5}$ $\leftarrow 0.151515 \dots$

$$\begin{aligned}
 0.\dot{1}\dot{5} &= 0.15 + 0.0015 + 0.000015 + \dots && \leftarrow \begin{array}{l} \text{初項 } 0.15 \\ \text{公比 } 0.01 \end{array} \\
 &= \frac{0.15}{1-0.01} \\
 &= \frac{0.15}{0.99} \\
 &= \frac{5}{33} \text{ ,,}
 \end{aligned}$$

(2) $0.1\dot{2}3\dot{4}$ $\leftarrow 0.1234234 \dots$

$$\begin{aligned}
 0.1\dot{2}3\dot{4} &= 0.1 + \underbrace{0.0234 + 0.0000234 + \dots} && \leftarrow \begin{array}{l} \text{初項 } 0.0234 \\ \text{公比 } 0.001 \end{array} \\
 &= 0.1 + \frac{0.0234}{1-0.001} \\
 &= 0.1 + \frac{0.0234}{0.999} \\
 &= \frac{1}{10} + \frac{234}{9990} \\
 &= \frac{137}{1110} \text{ ,,}
 \end{aligned}$$